

SAAD AIT YAHIA

Software Engineer — Systems Programming | Operating Systems | C/C++ | Linux

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PROFESSIONAL SUMMARY

Software engineering student with hands-on experience in designing Linux operating systems, a process/memory management simulator, and desktop applications in C/GTK, complemented by full-stack development and applied cybersecurity work. Comfortable from low-level concepts (kernel, system calls) up to backend services and secure authentication. Google-certified in IT Support and Cybersecurity; active contributor to the university's IT infrastructure and to open-source Linux tools. Seeking an internship in Software Engineering / Systems / C++.

TECHNICAL SKILLS

Areas of Interest: Operating Systems, Systems Programming, Linux, Computer Architecture, Cybersecurity

Certifications: Google IT Support Professional Certificate · Google Cybersecurity Professional Certificate

Programming Languages: C, C++, Python, Java, JavaScript, Bash / Shell Scripting

Systems & Operating Systems: Linux (Debian-based), Kernel Fundamentals, Process Management, Virtual Memory, System Calls, Multithreading, Package Management, Shell Scripting

Desktop / GUI Programming: GTK, Desktop Application Development in C

Backend & Frameworks: Flask, Django, Node.js, REST APIs, Bootstrap

Databases: MySQL, PostgreSQL, MongoDB, SQLite

Cybersecurity & Networking: Network Security Fundamentals, Authentication & Session Security, System Hardening, Troubleshooting

Development Tools: Git, GitHub, Docker, JIRA

PROJECTS

Highlights: 80+ public GitHub repositories, 40+ cumulative stars, OS/Linux, GTK and web projects.

univOS & FSBMOS — Linux-Based Operating Systems — *C, Linux, Bash, Kernel/OS Development — saad-programmer.github.io/univOS*

- Designed two lightweight Debian-based Linux operating systems (univOS, FSBMOS), derived from the MiniOS project, offered in four editions (Standard, Development, Advanced, custom university edition) for 32-bit and 64-bit architectures.
- Contributed to the system architecture and kernel-level operations, optimized system utilities and resource usage, and set up package management tools, persistent storage, and security patches.
- Collaborated with a university AI/Dev club and an open-source Linux distribution team to deliver and maintain the project.

OPYS — Operating System Simulator — *Python, Systems Design*

- Designed and developed a Python simulator modeling the internal mechanisms of an operating system, including process management, the file system, and security mechanisms.
- Implemented virtual memory management and system call emulation to demonstrate the low-level behavior of an OS in a controlled, testable environment.

GTK Desktop Application Suite — *C, GTK — github.com/Saad-programmer/GTKFinalVers, /AquariumGTK*

- Developed several native desktop applications in C/GTK (including an interactive aquarium simulation and various tools), covering event-driven interface design, widget composition, and memory management in C.
- Hands-on practice with low-level resource management and debugging in a compiled environment, outside the web.

EXPERIENCE

Web Developer & Infrastructure Contributor | *IT Tech Juniors (University AI & Systems Club)* 2024 – Present

- Contributed to a university club dedicated to AI, system administration, and IT infrastructure (15+ active members); organized technical workshops on Linux, networking, and ethical hacking.
- Supported ongoing improvements to the university's IT infrastructure, including security hardening and system optimization.

IT Support & Technical Assistant | *FSBM University (Faculty of Sciences Ben M'Sick)* 2023 – Present

- Served as a stand-in technician, resolving hardware, software, and network issues for professors, staff, and students, and supporting the IT infrastructure during university events.
- Wrote internal documentation for software installations, system configurations, and troubleshooting procedures used by fellow student technicians.

Cybersecurity & IT Support | *Independent / Freelance* 2023 – Present

- Earned the Google IT Support and Google Cybersecurity Professional Certificates; applied networking and security principles to freelance troubleshooting for clients.

EDUCATION

Engineering Degree — Software Engineering & Computer Systems Integration (ILISI) 2025 – 2028
Faculty of Sciences and Techniques of Mohammedia (FSTM), Hassan II University

DEUG — Mathematics, Computer Science and Physics (MIP) 2023 – 2025
Faculty of Sciences Ben M'Sick (FSBM), Hassan II University